

1. Which algorithm performs the fewest comparisons on already sorted data?
Both bubble sort (improved) and insertion sort got the fewest comparisons for my data set (24 comparisons). For simplicity, insertion sort is preferred.
2. Which algorithm performs the worst on reverse-ordered data?
Bubble sort, both original and improved, consistently has the worst performance on reverse-ordered data.
3. Which algorithms are most sensitive to the initial order of the data?
Bubble sort and insertion sort are the most sensitive to the initial order of the data due to shifts and comparisons resetting to first of the set for comparison.
4. How does the “improved” bubble sort behave differently from the regular bubble sort on sorted input?
The improved bubble sort behaves differently because it checks whether a swap has been performed. By checking if no swaps have been performed, it allows the algorithm to end early, as 0 swaps = sorted array.
5. Which algorithm appears most consistent regardless of input order
Selection sort appears the most consistent. Selection always compares the remaining unsorted portion, without any quick-exit logic.